Éric M. Savin

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Professional experience

2023–	Expert | Commission des titres d'ingénieur (CTI), Paris – France.
2020–	Expert advisor | Blue Spirit Aero (BSA), Rueil-Malmaison – France.
2018–	Research scientist | Computer Science Dept., ONERA, Palaiseau – France.
2013–2017	Research scientist | Fluid Mechanics Dept., ONERA, Châtillon – France.
2010–	Senior expert, mechanical & civil engineering | Ministry of Higher Education & Research – France.
2000–2013	Research scientist | Structural Mechanics Dept., ONERA, Châtillon – France.
1996–1999	Associate scientist | Centrale Recherche S.A., Châtenay-Malabry – France.
1994–1996	Consultant | Systra Group, Seoul – South Korea.
1994	Structural engineer | Systra Group, Paris – France.

Research interests: Structural dynamics & acoustics, Waves, Sound, Imaging, Computational physics, Uncertainty quantification, Data science.

Teaching appointments

2011–	*CentraleSupélec*	Professor (part time) | Engineering Mechanics Dept. Co-head (2011–2014), Graduate Program Materials & Structural Dynamics.
2002–2011	*École Centrale Paris*	Assistant Professor (part time) | Engineering Mechanics Dept.
1998–1999	*École Centrale Paris*	Teaching Assistant | Mathematics & Computer Science Dept.

Courses taught: Advanced structural dynamics & acoustics (2002–present, graduate), Continuum mechanics (2002–present, undergraduate), Structural dynamics & acoustics (2008–2015, undergraduate), Stochastic mechanics (2011–2014, graduate), Finite elements (2002–2008, undergraduate), Scientific computing (1998–1999, undergraduate).

Education & degrees

2010	*Sorbonne University*	Habilitation degree | Referees: M. Campillo, W. Desmet, R. Langley.
1996–1999	*École Centrale Paris*	Ph.D. Computational Mechanics | Advisor: D. Aubry.
1993–1994	*École Centrale Paris*	M.Sc. Structural Dynamics & Acoustics.
1990–1993	*École Centrale Lyon*	M.Eng. Civil Engineering | Minor in business administration at EM-Lyon.

Mentoring experience

Postdoctoral associates

1. Lorenzo Scaglione 2026-2027: *From diffusion to localization in strongly-scattering elastic materials* | Co-supervisor Régis Cottereau | Funded by ANR 82 k€.
2. C. Bocquet 2025: *Reduced bases for wave equations* | Co-supervisors Xavier Juvigny, Mathieu Lorteau, Jérôme Simon | Funded by ONERA 57 k€.
3. Y. Le Guennec 2012–2013: *Time reversal in beam trusses* | Funded by LMSS-Mat 50 k€.
4. I. Baydoun 2012–2013: *Anisotropic transport and diffusion of elastic waves* | Funded by CNRS 50 k€.

Ph.D. students (on-going: 2)

5. Adrian Padilla-Segarra 2023–2026: *Solving PDEs with physics-informed Gaussian process regression* | Paul-Sabatier University, co-advisors Profs. Pascal Noble & Olivier Roustant (INSA Toulouse) | Funded by ONERA & INSA Toulouse.
6. W. Genuist 2023–2026: *Solving PDEs with generative diffusion models* | Paris-Saclay University, co-advisors Profs. Didier Clouteau & Filippo Gatti (CentraleSupélec) | Funded by ONERA & Paris-Saclay University.

7. L. Bonnet 2018–2021: *Robust certification of aerodynamic design by concentration-of-measure inequalities* | Paris-Saclay University, co-advisor Prof. H. Owhadi (California Institute of Technology) | Defended Dec 13 2021 | Funded by French Ministry of Education 82 k€ | YouTube.
8. É. Gay 2015–2018: *Coherent interferometric imaging in fluid dynamics* | Paris-Diderot University, co-advisor Prof. J. Garnier (École polytechnique) | Defended Sep 19 2019 | Funded by DGA & ONERA 60 k€.
9. Y. Le Guennec 2009–2012: *Transient dynamics of beam trusses under impulse loads* | École Centrale Paris, co-advisor Prof. D. Clouteau (CentraleSupélec) | Defended Feb 4 2013 with honors | Funded by ONERA 60 k€.
10. J. Staudacher 2007–2010: *Conservative numerical schemes for high-frequency wave propagation in heterogeneous media* | École Centrale Paris, co-advisor Prof. D. Clouteau (CentraleSupélec) | Defended Nov 6 2013 with honors | Funded by CNES 48 k€.
11. S. Bougacha 2004–2008: *High-frequency oscillations in bounded elastic media* | Évry-Val-d'Essonne University, advisor Prof. R. Alexandre (French Naval Academy) | Defended Jan 14 2010 with honors | Funded by CNES 48 k€.

Undergraduate students: principal advisor of 13 master's theses, and 3 student projects at École Centrale Paris, University of Cambridge, ESSEC Business School.

Synergistic activities

Services

- Board member (nominated 2020–), Graduate School of Mathematics, Paris-Saclay University.
- Board member (nominated 2020–2026), Institute of Aeronautics and Astronautics, Paris-Saclay University.
- Board member (nominated 2018–2023), GdR MécaWave, CNRS.
- Board member (elected 2020–2022), Social and Economic Committee, ONERA.

Main research grants

- ANR: Localisation des ondes pour la réduction des émissions acoustiques large-bande, 2025–2028 (634 k€; co-PI).
- ONERA: Mesoscopic passive imaging with acoustic, elastic and electromagnetic waves, 2015–2016 (623 k€; PI).
- European Union: Uncertainty management for robust industrial design in aeronautics, 2013–2016 (3.90 M€).
- CNRS F2M: Anisotropic transport and diffusion of elastic waves, 2012–2013 (50 k€; PI).
- ANR: Calcul intensif stochastique et sûreté des systèmes industriels, 2007–2010 (397 k€).

Refereeing

- Editorial board: *Interaction Multiscale Mech.* (2007–2014).
- Scientific board: *CSMA Nat. Coll.* (2006–2014), *Int. Conf. Adv. Interaction Multiscale Mech.* (2009–2014).
- Review panel: Czech Science Foundation (Czech Republic), EU's Horizon 2020 programme (Brussels), MESRI (France), Inria (France), INSA Lyon (France).
- International journals (*ca.* 150).

Recent invited talks

1. Workshop "Mathematical and computational foundations of digital twins" (CIRM, Marseille, France, Aug 11-15 2025; slides).
2. Workshop "Digital twins for inverse problems in Earth science" (CIRM, Marseille, France, Jul 22-26 2024; slides).
3. JOSO 2019, Institut Lasers & Plasmas (Le Barp, France, Mar 13 2019; slides).
4. Workshop "New mathematics for a safer world: wave propagation in heterogeneous materials" (Edinburgh, Scotland, June 12-15 2017; video).
5. 14th Int. Conf. Zaragoza-Pau on Mathematics and its Applications (Jaca, Spain, Sep 12-15 2016; paper).

Recent publications

1. J.-L. Akian, É. Savin, *Wave Motion* **127**:103296 (2024).
2. J.-L. Akian, L. Bonnet, H. Owhadi, É. Savin, *J. Comput. Phys.* **470**:111595 (2022).
3. J.-L. Akian, É. Savin, *SIAM Multiscale Model. Simul.* **19**(3):1394-1424 (2021).
4. É. Gay, L. Bonnet, C. Peyret, É. Savin, J. Garnier, *J. Sound Vib.* **494**:115852 (2021).
5. J. Garnier, É. Gay, É. Savin, *SIAM Multiscale Model. Simul.* **18**(2):798-823 (2020).